

Temperate forest

Temperate forest grows in regions that have a wide range of climates. In some, winters are cold and summers are cool; in others, the winter is relatively mild, and the summer heat rivals that in the tropics. Where winters are cold, temperate forest trees are usually deciduous, shedding their leaves in winter and growing a new set in spring; in warmer regions, many trees keep their leaves all year. Although temperate forest does not have as many animal species as tropical forest, it is still among the richest wildlife habitats on land.

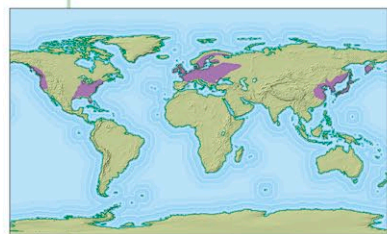
Deciduous forest

In the depths of winter, deciduous forest can seem gaunt and empty, and largely devoid of animal life. But as the days lengthen in spring, and buds begin to burst, the habitat becomes alive with birdsong and animals on the move. This transformation is triggered by a sudden abundance of plant food—one that nourishes large numbers of both the plant-eating insects as well as the animals that feed on them. Many of these forest animals are permanent residents, but they also include migratory birds that fly in from distant parts of the world.

Compared with tropical forest, temperate deciduous forest has relatively few tree species: the maximum number—found in some of the forests of eastern North America—is several hundred, while tropical forest might contain several thousand. Nevertheless, temperate forest trees are powerhouses of life. Large oak trees, for example, can produce over a quarter of a million leaves a year—enough to sustain the army of weevils, gall wasps, and moth caterpillars that feed rapidly in spring and early summer while the leaf crop is at its freshest and most nutritious.

Like tropical forest, deciduous forest has a clear, vertically layered structure, but there are some important differences. The trees are

rarely more than 100ft (30m) tall, and the canopy layer is usually deep but open, allowing light to reach the understory and encourage plant growth. Fallen leaves rot slowly in cool conditions, so deciduous forest has an unusually deep layer of leaf litter that insects, woodlice, and millipedes use as food and cover. This means that while many small animals live in the cracks and crevices in bark, the place that is richest in invertebrate life is not the trees but the ground.



LARGE AREAS of temperate deciduous forest are found in the Northern Hemisphere; most temperate forest in the Southern Hemisphere is evergreen.



MOTHS lay their eggs on leaves, buds, or bark. Their caterpillars are often the most numerous leaf eaters in the forest.



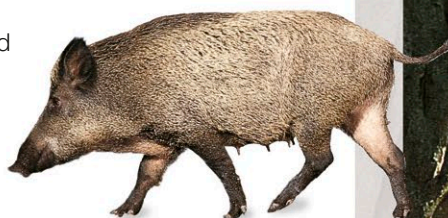
TREECREEPERS patrol tree trunks in search of small insects hidden in bark crevices.



SQUIRRELS spend the autumn collecting food, which they then store away for use in the winter.



DEER feed on leaves in summer; in winter, they often strip bark from shrubs and young understory trees.



WILD BOAR dig up the leaf litter with their snouts, feeding on animals, nuts, and roots.



IN THE MOIST conditions of the forest floor, lungless salamanders can absorb oxygen through their skin.

CANOPY

UNDERSTORY

GROUND

